Genetic counseling and testing practices for late-onset neurodegenerative disease: a systematic review

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Online resource 3 *Detailed summary of included papers*

Study	Paper reference and author location	Condition(s) investigated[#]	Objectives	Sample characteristics*	Study type	Testing type	Main findings relevant to the systematic review	Genetic counseling goal(s) addressed	Qualsyst result (Total score/ total possible score) and quality issues
	Genetic counseling practice sourced from clinical experience								
1	Eno et al. (2020) [97] USA	HD	Report the use of the electronic health record (EHR) for presymptomatic HD GT across different HD Centers (i.e. how the HD gene analysis is ordered, resulted and stored)	23 clinical care teams (HDSA Centers of Excellence) (53% response rate)	Cross-sectional survey	Predictive	Most teams have developed their own practices and there was much variation in whether the following were recorded in the EHR: encounters, notes, results, GT ordered, and whether pathology received by laboratory	Education	0.93 (13/14) Question/objective not clearly described
2	Bardakjian et al. (2019) [70] USA	HD	Report the experience of a new HD Center since its inception, estimating the capture of the population served, describing the care provided and measuring changes in client behavior in response to release of research-related information	266 unique HD clients seen in HD Center 145 seen in 2018: 88 with manifest HD, 28 premanifest mutation carriers, 12 who underwent predictive GT and did not carry an expanded allele, 8 who requested but	Retrospective case series	Diagnostic Predictive	New center demonstrated high demand for in-clinic multidisciplinary care Neurologist and nurse involved in 100% of encounters, followed by psychiatrist and genetic counselor Demand for predictive GT significantly increased following	Interpretation Education Counseling	0.92 (11/12) Results not sufficiently described

				did not complete predictive GT, 7 who decided against predictive GT, 2 phenocopies with negative GT			the press announcement of successful completion of clinical trial		
3	Bonnard et al. (2019) [56] France	HD	Compare the age, motivations, and time required before deciding to have GT performed between 25% at-risk and 50% at-risk individuals Compare outcomes in 25% at-risk individuals, between those who are variant positive or variant negative Observe whether revealing the parent's status adversely affected the parent–child relationship Understand the familial context that led them to request GT before their at-risk parent	1611 individuals requested predictive GT between 1992-2016 1456 at 50% risk: 73% underwent predictive GT 155 at 25% risk: 60% underwent GT, 14/94 were variant positive 9 variant positive individuals at 25% risk and 9 age-matched variant negative individuals. Four participated in further semi-structured interview (2/4 reported as case studies).	Consecutive case series Qualitative study	Predictive	Most common motivation in 50% risk and 25% risk group was "to know" Four adverse reactions when individual at 25% risk underwent GT and informed intervening relative: 3 intervening relatives became depressed, 1 suicide 1 month after the result had been disclosed	Interpretation Education Counseling Support	0.60 (18/30) Quantitative results well described Qualitative study design, and data collection not clearly described Theoretical framework/ wider body of knowledge, data analysis, verification procedures and reflexivity of the account not described Conclusions not well supported
4	Crook et al. (2019) [100] Australia	ALS	Present the case of an ALS patient who underwent GT through our motor neurone disease clinic Highlight current limitations to analysing and interpreting C9orf72 expansion GT results and describe how this resulted in discordant reports of pathogenicity between GT laboratories that confounded the GC process	1 ALS patient who received discordant C9orf72 expansion results and interpretation	Case study	Diagnostic Predictive Reproductive	Discordant results confounded GC process, highlighted difficulties in GT and interpreting C9orf72 results Discordant results have associated potential psychological and legal risks for the client and health provider	Interpretation Education	0.90 (9/10) Question/ objective not clearly described
5	Klepek et al. (2019) [72]	ALS	Characterize clinician practices regarding GT and GC, perceived	80 ALS clinicians and members of the Northeast ALS	Cross-sectional survey	Diagnostic	Lack of consensus in ALS GT practices: 92.3% offered GT to	Interpretation Education	1.00 (16/16)

	USA		challenges, and attitudes and other factors that may be associated with the offer of ALS GT Compare clinician attitudes towards GT to attitudes of persons with ALS	Consortium (response rate 31.4%) 96.2% were neurologists			patients with familial ALS, 57% to sporadic ALS with family history of dementia, 36.9% to sporadic ALS Divergent views between clinicians and patients: clinicians less likely to have GT themselves, or see value in GT for relatives		
6	Paneque et al. (2019) [33] Portugal	LONDs including: HD TTR-FAP SCA3, SCA2, SCA7 CADASIL (dentatorubral-pallidoluysian atrophy)	To describe the profile of the population seeking presymptomatic GT, while also reflecting on the experience and conducting the protocol of multidisciplinary sessions since 1996	1498 requested predictive GT, 240 withdrew, 28 were excluded 1230 underwent predictive GT, 680 non-carriers, 550 carriers	Retrospective cohort study	Predictive	45% did not follow up after receiving GT results, 29.6% were seen a year post-GT Most common reason for GT to reduce uncertainty (41.7%)	Education Counseling Support	1.00 (22/22)
7	Tibben et al. (2019) [94] The Netherlands	HD	Describe four cases in which the couples and clinicians involved were confronted with an unexpected outcome of prenatal GT	4 couples in which expanded CAG repeats were observed in (or stemming from) the presumed non-HD side	Non-consecutive case series	Predictive Reproductive	Population risks of HD should be a required discussion to ensure comprehensively informed reproductive GT and GC	Interpretation Education	1.00 (10/10)
8	Olszewska et al. (2018) [101] Ireland	>6 LONDs	Perform a retrospective chart review/ cohort analysis of the Neurogenetics clinic over 12 months, reviewing symptoms and work up data	27 individuals who attended a pilot neurogenetics clinic	Consecutive case series	Diagnostic	Benefits of multidisciplinary team to address gap in service delivery Identification of pathogenic variants directed screening, treatment and facilitated onward GC	Interpretation Education Counseling Support	0.80 (8/10) Question/ objective not clearly described Conclusions not well supported
9	Charles et al. (2017) [68] USA	HD	Highlight the difficulties involved with care of an extended family with HD living on a small island nation due to their low socioeconomic status,	1 family with HD who live on several resource-limited Caribbean Islands	Case study	Diagnostic Predictive	Genetic and clinical diagnosis can be impeded by lack of resources and lack of access to specialty care	Interpretation Education Support	1.00 (10/10)

			barriers to accessing medical care and geographical isolation				Definitive diagnosis positively impacted family by facilitating GC, community outreach, and dispelling disease myths.		
10	Goldman et al. (2017) [55] USA	HD ALS/FTD	Demonstrate the complex nature of GC and GT in the presence of psychiatric symptoms, whether emanating from the disease itself or the results of living in an affected family	4 individuals with psychiatric symptoms who requested predictive GT	Non-consecutive case series	Predictive	Psychiatric symptoms may emanate from the disease itself, or living in an affected family Health providers must still prepare clients for positive and negative results Protocol may need to proceed slowly to foster positive outcome	Interpretation Education Counseling support	1.00 (10/10)
11	Mandich et al. (2017) [46] Italy	HD	Report the sociodemographic characteristics of predictive GT applicants, their motivations and expectations, and the outcomes of the GC protocol during two decades of direct HD GT.	299 individuals who applied for predictive GT between 1993-2014	Retrospective cohort study	Predictive	Protocols completed more in men (68.5% vs 53.5%), those over 25 (63.4% vs 48.1%) Factors influencing the decision-making process differed between males and females	Counseling Support	0.82 (18/22) Analytic methods, results and likely confounders not sufficiently described
12	Mantero et al. (2017) [75] Italy	ALS/FTD (Parkinson's disease)	Discuss the issues that arose in family GC for likely sporadic ALS and parkinsonism-dementia complex (ALS-PDC)	1 individual with likely sporadic ALS-PDC	Case study	Diagnostic	GC important for family even if low recurrence risk, to provide support and discuss limitations	Interpretation Education Counseling	1.00 (10/10)
13	Vajda et al. (2017) [99] Ireland/ UK	ALS	Determine the degree of consensus among clinicians on the clinical use of GT in ALS and the factors that determine decision-making	167 ALS clinicians from 21 different countries 86.8% were neurologists	Cross-sectional survey	Diagnostic Predictive	90.2% offer GT to patients defined as having familial ALS, 49.4% to sporadic ALS 42% never off presymptomatic GT Responses varied between ALS specialists and non-specialists and based	Interpretation	1.00 (16/16)

							on number of new patients seen		
14	Clift et al. (2016) [54] USA	Prion disease	Present an example case which discusses the psychosocial issues encountered and the role of GC in presymptomatic GT for incurable neurodegenerative conditions	1 individual who sought predictive GT after familial CJD was confirmed in her mother on post-mortem GT	Case study	Diagnostic Predictive	Multidisciplinary approach key to GC care for the family Clinicians should be aware of GC resources	Interpretation Education Counseling Support	0.90 (9/10) Question/ objective not clearly described
15	Stark et al. (2016) [76] Australia	Unspecified LOND	Explore a complex case where the GT wish of one family member was in direct conflict to that of another, assess the potential benefits and harms from acceding to or denying such a request, and present an approach to balancing competing rights of individuals within families	1 family at risk of an unspecified LOND: the client at 25% risk requested predictive GT, the intervening relative did not wish to know and would commit suicide if mutation positive	Case study	Predictive Reproductive	Approach described to balancing competing rights in a family Magnitude of risks for client and relatives should be considered and every effort made to limit adverse outcomes in GT process	Interpretation Education Counseling	1.00 (10/10)
16	Clement et al. (2015) [49] France	HD	Review the historical context of guidelines and good clinical practice, the experiences of our team covering more than 20 years of predictive GT for HD in France, and the new French legislation, all factors that regulate presymptomatic GT	1705 persons at risk of HD who requested GC between 1992 and 2013.	Consecutive case series	Predictive	47% withdrew from predictive GT protocol demonstrating that request for GT does not imply client wants to know New legislation of health providers to disclose family medical information may impact on predictive GT uptake, due to concerns about confidentiality Benefit of a multidisciplinary approach	Interpretation Education Counseling Support	0.70 (7/10) Subject characteristics and results not sufficiently described Conclusions not well supported
17	Cruz-Marino et al. (2015) [57] Cuba	SCA2	Review the 13-year experience of the SCA2 predictive GT program in Cuba, describing different ethical, psychosocial, and technical challenges that	1193 individuals who requested predictive GT within a 13-year period	Consecutive case series	Predictive Reproductive	895 completed the protocol: 43.4% uptake of predictive GT, 23.9% uptake of reproductive GT (10/33 couples carried	Interpretation Education Counseling Support	1.00 (10/10)

			led to major changes in the predictive GT protocol				test-positive fetus to term) Some withdrew due to protocol length Benefit of multidisciplinary team and practices at the community level demonstrated		
	Cruz-Marino et al. (2013) [45] Cuba	SCA2	Review the 11-year experience of predictive GT for SCA2, including the pre-GT opinions about different aspects of the protocol and the profile of at-risk individuals who underwent GT	1050 individuals who requested predictive GT between 2001-2011	Consecutive case series	Predictive	768 completed the protocol, predictive GT uptake 24.91% 31 symptomatic individuals were eliminated from the predictive GT protocol at their first appointment (neurological assessment)	Interpretation Education Counseling Support	1.00 (10/10)
18	Mandich et al. (2015) [74] Italy	ALS	Report several issues in GC for ALS	2 siblings with ALS who had discordant GT results	Case study	Diagnostic Predictive	One sibling may have phenocopy Benefits highlighted around exploring the complexity and pitfalls of GT and GC, the unexpected consequences for relatives pre-GT, and using multidisciplinary team	Education Counseling	1.00 (10/10)
19	Klitzman et al. (2014) [29] USA	HD AD (Tay Sachs, CF, autism, sex selection)	Survey attitudes and practices to understand whether providers in neurology and psychiatry discuss PND and PGD with clients, and if so, how frequently, when, how and what factors are involved	535 health providers: 163 neurologists 372 psychiatrists	Cross-sectional survey	Reproductive	24.9% of neurologists, and 31.9% of psychiatrists had discussed PND 95.3% didn't feel comfortable discussing PGD	Education	1.00 (18/18)
20	Schuler-Faccini et al. (2014) [44]	SCA3	Present our experience from two programs conducting predictive GT for SCA3 in Porto, Portugal and Porto	329 individuals who sought predictive GT for SCA3, 263 from Brazil	Consecutive case series	Predictive	50% from Brazil, 77% from Portugal underwent GT	Interpretation Education Counseling Support	1.00 (10/10)

	Brazil, Portugal		Alegre, Brazil from 1999-2012 Report an illustrative GC case	1 family case example	Case study		Benefit of multidisciplinary team demonstrated		
21	Smith et al. (2014) [59] USA	ALS HD	Describe the challenges and lessons learned from a case in which an individual with a fatal condition was at risk for a second fatal condition and had difficulties with communication	1 individual with ALS at 50% risk of HD	Case study	Diagnostic Predictive	GC challenges inherent in this case: difficulty communicating due to disease progression, diagnostic consideration of two fatal conditions, complex risk information, personal and familial implications	Interpretation Education Counseling	1.00 (10/10)
22	Tanaka et al. (2013) [102] Japan	>6 LONDs	Present the results of a follow-up nationwide survey on predictive GT for LONDs in Japan	60 institutional members of Japan's National Liaison Council for Clinical Sections or Medical Genetics (response rate 67.4%)	Cross-sectional survey	Predictive	301 clients interested in predictive GT over 5 year period, 93 underwent GT Lack of non-MD counseling staff was apparent, clinical geneticists predominantly involved	Counseling Support	1.00 (14/14)
23	Gonzalez et al. (2012) [38] Portugal	SCA3	Assess the following in individuals who had 5 years prior received positive results from predictive GT for SCA3: the psychological well-being, family satisfaction/ occurrence of familial changes, and the role played by a number of factors, such as presence of symptoms, in general psychological wellbeing and family satisfaction	47 individuals from the Azores archipelago who had positive predictive GT results approximately 5 years prior and attended their 4th post-GT psychological evaluation session, representing nearly 80% of the total number of individuals who tested positive	Retrospective cohort study	Predictive	More than half demonstrated moderate (28.9%) or severe (23.7%) stress Most (59.6%) had high familial satisfaction Development of first symptoms negatively impacted psychological state	Counseling Support	0.95 (21/22) Conclusions not well supported
24	Reyes et al. (2012) [27] France	CADASIL	Analyse the profiles and motivations of individuals at risk of CADASIL who	33 individuals who requested predictive GT	Consecutive case series	Predictive	63% dropped out pre-GT	Education Counseling Support	1.00 (20/20)

			requested predictive GT between 2003-2010 Identify the neurological, cognitive and psychological modifications observed in applicants who received a positive result	 11 completed neuro and psychological examination after receiving results and 18 months later	 Prospective cohort		High overall quality of life reported in those who could be followed up Multidisciplinary and multistep practice through to protect from harm		
25	Van Rij et al. (2012) [58] The Netherlands Belgium France	HD	Provide a comparative overview of PGD approaches and technical workup for HD between 1995-2008 across 3 European centers Study differences in the populations who apply for PGD and their reproductive histories Compare PGD results between the centers and compare them with literature data	331 couples received GC	Prospective cohort study	Reproductive	68% requested direct PGD GT, 32% requested exclusion PGD 257 started PGD workup, 29 were rejected, 61 refrained from PGD Overall delivery rate of couples starting ≥1 PGD cycle 37.4% (65/174)	Interpretation Education	1.00 (22/22)
26	Yanoov-Sharav et al. (2012) [105] Israel	FSHD	Present our experience of GT and GC for FSHD between 2000 -2006 Present a case study which highlights a unique example of GC for dominant, relatively late-onset disease	66 individuals who underwent GT for FSHD (59 diagnostic GT, 7 predictive GT) 1 family case example	Consecutive case series Case study	Diagnostic Predictive	< 60% received pre-GT GC, <30% received post-GT GC Pre-GT GC and multidisciplinary care emphasized due to complexities of LONDs and molecular GT	Interpretation Education Counseling	1.00 (10/10)
27	Dufrasne et al. (2011) [41] Canada	HD	Report and analyse the uptake, reasons given for requesting predictive GT, social and demographic characteristics, GT outcomes, and emotional reactions of individuals who proceeded with HD predictive GT and explore how best to fulfill participants' perceived needs	181 individuals who requested predictive GT between 1994 and 2008	Consecutive case series	Predictive Reproductive	135 completed GT >1 reason for predictive GT usually mentioned, most common eliminating uncertainty Prenatal GT not frequently requested	Interpretation Education Counseling Support	1.00 (18/18) Described as retrospective cohort study

28	Cruz Marino et al. (2011) [39] Cuba	SCA2 (Friedrich's Ataxia)	Describe some of the ethical dilemmas that arose in predictive GT for hereditary ataxias in Cuba Explore the GC process and the decisions made during predictive GT and prenatal diagnosis	4 case examples with ethical dilemmas: identical twins, GT an individual at 12.5% risk, GT a foetus at 25% risk and misattributed paternity	Non-consecutive case series	Predictive Reproductive	Complexities of predictive GT are apparent and expanded guidelines required to address these ethical issues	Interpretation Education Counseling Support	1.00 (12/12)
29	Butler et al. (2011) [62] Canada	AD	Identify GC challenges and describe our specific GC approach for members of a geographically remote and culturally distinct community	1 family with early-onset familial Alzheimer disease (EOFAD) caused by PSEN1 in a North American Aboriginal community	Case study	Diagnostic Predictive	Alternative approaches to disseminating genetic information and ensuring appropriate, confidential and accessible GC services described	Interpretation Education Counseling Support	0.80 (8/10) Question/ objective and study design not clearly described
30	Futter et al. (2009) [67] South Africa	HD	Compile a comprehensive profile of the participants who had undergone predictive GT for HD in the West Cape region of South Africa to inform changes to improve GC services	36 individuals who had undergone predictive GT between 1995-2005 27 participated in interviews	Consecutive Case series Qualitative study not reported	Predictive	Uptake of GT in those with mixed ancestry was significantly lower Possible barriers: limited access to GT due to low income or education	Education	0.86 (12/14) Only quantitative data assessed Analytic methods and results not sufficiently described
31	Riedijk et al. (2009) [63] The Netherlands	FTD	Unclear. Assumed objectives: To report predictive GT uptake and outcomes between 1999-2008 To present a case study and propose the idea of separation-individuation	100-180 individuals from familial FTD families and at 50% risk	Consecutive case series Case study	Predictive	13 requested GC between 1999 and 2002, 13 underwent GC between 2003 and 2008 , 1 underwent PND Low acceptance of GT hypothesized due to theoretical framework of separation-individuation	Education Counseling Support	0.50 (5/10) Question/ objective and results not sufficiently described Study design not described Conclusions not well supported
	Genetic counseling practice trialed in clinical setting								
32	Spiers et al. (2020) [65] UK	HD	Evaluate participants' experience with a GC narrative group session to determine whether participating in a single GC narrative group is perceived as helpful	12 individuals who had tested positive on predictive GT and had participated in one of three GC group sessions between December	Qualitative	Predictive	Group had a positive impact of being able to meet and empathize with others in a similar situation, increased disclosure to others and improved mood and future outlook	Support	1.00 (20/20)

				2017 and March 2018					
	Stopford et al. (2020) [66] UK	HD	Explore presymptomatic individuals' (and their partners) experiences of a structured narrative group session to understand the value and feasibility of integrating narrative practices within a GC session	8 individuals who were purposively selected and attended a single narrative group sessions, 6 mutation positive, 2 male partners (not at risk of HD)	Qualitative	Predictive	Positive feedback received, highlighting importance of time and space for structured sharing of experiences	Support	1.00 (20/20)
	Macleod et al. (2018) [64] UK	HD	Explore the feasibility of offering narrative group sessions in the context of a predictive GT follow-up clinic Determine how participants experienced the session and whether they would recommend participation to others	9 individuals who had tested negative on predictive GT	Uncontrolled before and after study Qualitative	Predictive	Group sessions were seen as safe and enjoyable, and benefits included feeling less isolated, being inspired by other's stories and connecting as a group	Support	0.95 (38/40) Qualitative: analytic methods and reflexivity of the account not clearly described
33	Esplen et al. (2013) [34] Canada	HD (cancer, haemochro-matosis)	Develop a brief, reliable and valid instrument to screen psychosocial risk among those who are undergoing GT for adult-onset hereditary diseases	31 individuals from HD families undergoing GT participated (4% of total participants)	Prospective cohort	Unspecified	5 (23.8%) demonstrated distress 1 month post-GT Screening tool developed for further investigation in clinic or research	Counseling Support	0.95 (21/22) Likely confounders not described
34	Hawkins et al. (2013) [40] Canada	HD	Report the predictive GT undertaken between January 2011- January 2012 using telehealth and usual care Examine whether telehealth improves access to HD predictive GT while maintaining quality of care and support	41 individuals referred for predictive GT, 28 requested telehealth (15 attended at least one session, 14 completed survey, 10 received results, 8 completed 2nd survey) 13 who utilized usual care (11 received results and completed 2nd survey)	Prospective cohort	Predictive	No significant differences between individuals undergoing GT in person or by telehealth with respect to quality of care, information, counseling and support Majority were satisfied with GC process in both groups	Interpretation Education Counseling Support	0.91 (20/22) Question/ objective and subject characteristics not clearly described
35	Mariotti et al. (2010) [30]	HD SCA1	Define a well-framed, structured and easy procedure for GC in	92 individuals undergoing predictive GT GC	Consecutive case series	Predictive	72 (78%) proceeded with program, 55 (60%) received GT	Interpretation Education	1.00 (22/22)

	Italy	SCA2 SCA3 SCA17	subjects at risk for LONDs, in which psychological support is intended both for the client and the health provider Verify feasibility and effectiveness of this procedure and compare possible differences in the impact of predictive GT in subjects with SCAs and HD	60 at risk of HD, 32 at risk for SCAs			result, 38 (41%) completed entire program The need for psychological support was recognized for 5 mutation carriers and a non-carrier Clinical conference supported client and health provider	Counseling Support	
	Genetic counseling practice recommended from clinical research								
36	Oosterloo et al. (2020) [32] The Netherlands	HD	Provide an overview of the experiences of Dutch persons at risk of HD in consulting a neurologist before or after DNA analysis Make a recommendation if and at what moment in the GT procedure the judgment of a neurologist is desirable	71 individuals at risk of HD who visited one of 4 Dutch GC clinics, 32 saw a neurologist before GT, 12 after GT, 27 did not see a neurologist. 68 completed predictive GT (29% response rate)	Cross-sectional survey	Predictive	41/44 felt visit to neurologist was positive 59 desired consulting a neurologist, even those who did not have the gene expansion, suggesting consultation before GT may be beneficial	Interpretation Education	0.95 (19/20) Outcome measures partially reported
37	Schwartz et al. (2019) [104] France	Prion disease	Understand the feelings of at risk individuals towards predictive GT, their decision-making, and the long-term consequences Understand specific issues raised by PRNP-related disease	30 individuals who consulted the genetic department regarding predictive GT between 2004-2017 3/8 mutation carriers, 10/12 non-carriers and 6/10 who declined GT	Case series Qualitative (including psychological instruments)	Predictive	Anxiety rates high in non-carriers and untested subjects, highlighting psychological burden of living in a family with inherited prion disease	Counseling support	0.57 (17/30) Case series and qualitative study: question/ objective and study design not clearly described Qualitative study: data collection methods not clearly described. Theoretical framework/ wider body of knowledge, data analysis, verification procedures and reflexivity of the account not described

38	Ledo et al. (2018) [35] Portugal	HD TTR-FAP	Investigate long-term consequences of predictive GT to identify variables that may predict middle and long-term psychological disturbance due to predictive GT	196 individuals who had previously undergone predictive GT (28.6% response rate) 167 at risk of TTR-FAP, 29 at risk of HD	Prospective cohort	Predictive	Psychopathological assessment pre-GT can inform those who may need psychological support several years later Result of predictive GT not a relevant predictor	Counseling Support	1.00 (20/20)
39	Stuttgen et al. (2018) [36] USA	HD	Analyse long term changes in risk perception, and investigate factors that contributed to changes in risk perception	186 individuals who underwent predictive GT and had provided risk perception values before and after GT 39 had concurrent research clinic notes and semi-structured interviews, 27 referred to risk perception in interviews	Retrospective cohort Qualitative	Predictive	27% had unexpected changes in risk perception after GT results disclosure, particularly in those with repeat expansions Risk perception influenced by more than just results of predictive GT	Education Support	0.85 (34/40) Quantitative analytic methods and variance estimates not clearly described Theoretical framework and reflexivity of the account not described
	Stuttgen et al. (2018) [92] USA	HD	Examine opinions on the importance of autonomy in the decision to be tested for HD, whether a formal HD GT protocol is necessary, whether a physician ordering HD GT in the absence of a formal HD protocol is acceptable, whether ordering presymptomatic GT for HD online via a direct-to-consumer (DTC) website is acceptable, and whether incidental/ secondary findings of HD should be returned in the context of whole exome/ genome sequencing	39 recent interviews with individuals who underwent predictive GT between 1986-1998 15 expansion carriers, 21 non-carriers, 3 who dropped out before GT result disclosure	Qualitative	Predictive	Most supported individual's right to decide whether and when to pursue HD GT (31/38), use of a formal HD GT protocol (22/37), and returning medically actionable secondary findings (34/36) Most were opposed to physician ordering (28/35) and DTC HD GT (24/31) in the absence of a formal protocol and returning a secondary finding of an expanded HD allele (18/37)	Counseling	0.75 (15/20) Question/ Objective not clearly described Theoretical framework and reflexivity of the account not described
40	Ibisler et al. (2017) [60]	HD	Prospectively follow the decision-making process	72 individuals who participated in at	Prospective cohort	Predictive	93.4% had already sought information via	Education Counseling	0.86 (36/42)

	Germany		of individuals at risk in our center Explore their experiences following the decision as well as the impacts of GT results	least one of three surveys 31 participated in telephone interview	Qualitative		the internet before the first GC session More participants with an affected mother (56.9%) than an affected father (31%) sought GC	Support	Theoretical framework/ wider body of knowledge and verification procedures not described Data analysis and reflexivity of the account not clearly described
41	Leite et al. (2017) [96] Portugal	TTR-FAP HD SCA3 (haemochro-matosis)	Understand why subjects at-risk of LONDs want to undergo predictive GT Compare results with the motivations of subjects at-risk for hereditary haemochromatosis	213 individuals at risk of 3 LONDS (174 TTR-FAP, 34 HD, 5 SCA3)	Qualitative Case series (unclear whether consecutive or non-consecutive)	Predictive	Most common motivations: reasons related to the future, reasons related to others and curiosity and the need to know - all reasons external and unrelated to the characteristics of the disease	Education Counseling	0.93 (37/40) Quantitative and qualitative analytic methods, and reflexivity of the account not clearly described
	Leite et al. (2017) [53] Portugal	TTR-FAP HD SCA3 (haemochro-matosis)	Investigate what subjects at risk for TTR-FAP, HD, and SCA3 know about these 3 diseases in comparison with the knowledge that subjects at risk for HH have about the conditions	213 individuals at risk of 3 LONDS (174 TTR-FAP, 34 HD, 5 SCA3)	Qualitative	Predictive	References to the disease, references to the family and metaphors were mentioned more by subjects at risk of a LOND	Education	0.90 (18/20) Analytic methods and reflexivity of the account not clearly described
42	Quaid et al. (2017) [95] USA	HD	Examine factors associated with the decision of research participants who changed their minds and opted to undergo presymptomatic HD GT, compared with those who still chose not to be tested	1001 individuals at risk of HD who are a part of the PHAROS observational study, 104 underwent predictive GT after initially declining	Prospective cohort	Predictive	Baseline behavioral scores (especially apathy) were more strongly associated with later GT than motor and chorea scores Following GT, 56% of those who tested negative had less depression compared to prior, depression stayed the same or increased for 64% of those with repeat expansion	Counseling Support	1.00 (22/22)

43	Andersson et al. (2016) [28] Sweden	HD	Describe a couple's long-term experiences (from 6 months after result disclosure) and the consequences of predictive GT	1 couple interviewed separately on 9 occasions over a 2.5 year period	Qualitative - longitudinal descriptive case	Predictive	Long-term consequences of GT devastating for both members of the couple: anxiety, repeated suicide attempts, financial difficulties and divorce Long-term support recommended for both client and partner	Counseling Support	1.00 (20/20)
	Andersson et al. (2013) [26] Sweden	HD	Describe the prospective experience of a client undergoing a presymptomatic GT for HD, and her husband in order to obtain an understanding of the client's perspective and the effect on the couple	1 couple interviewed separately on 9 occasions over a 15 month period	Qualitative – longitudinal descriptive case	Predictive	Throughout pre- and post-GT, need to acknowledge needs of client and partner, particularly important at time of results and symptom onset	Education Counseling Support	1.00 (20/20)
44	Benatar et al. (2016) [52] USA	ALS	Highlight clinically relevant aspects of the genetic complexity of ALS and present an approach to predictive GT that we have developed and refined over the last 8 years in the pre-FALS study	317 GC sessions with 161 individuals at 50% risk of familial ALS who are part of the pre-fALS study 75 post-GT sessions with 63 individuals with familial ALS	Consecutive case series	Predictive	Clients may be interested in research participation without results being disclosed Pre-GT GC requires detailed discussion and careful consideration of potential pitfalls of proceeding	Interpretation Education Counseling Support	0.70 (7/10) Study design and results not sufficiently described Conclusions not well supported
	Fanos et al. (2011) [71] USA	ALS	Explore the basis for participants' decision to learn results of presymptomatic GT or not, understand the psychosocial impact of the decision and assess attitudes toward receiving results by telephone or in person	20 individuals at 50% risk of familial ALS, who are part of the pre-fALS study 14 elected to receive results (8 mutation carriers, 6 non-carriers)	Qualitative	Predictive	Telephone counseling as option for those who can't easily access in person counseling Those who decline GT may change their mind in future Clients adapted well in the short-term	Education Counseling Support	0.80 (16/20) Theoretical framework and reflexivity of the account not described
45	Paneque et al. (2015) [89] Portugal	All LONDs tested for in Portugal	Explore professionals' views of relevant quality indicators in their own GC practice concerning predictive GT for LONDs	18 genetic health professionals (85% of total eligible interviewees)	Qualitative	Predictive	Core components of GC identified Challenges specific to LONDs: ambiguity of health/ illness status,	Interpretation Education Counseling Support	0.90 (18/20) Reflexivity of the account not described

			Examine current assessment of such GC practice in Portuguese genetic services				time burden for health professionals Health professionals associated quality with non-directiveness, information given and comprehension pre-GT, decision-making facilitated based on consultands' motivations		
46	Guimaraes et al. (2013) [42] Portugal	TTR-FAP HD SCA3	From the client's perspective, recognize aspects relevant across the predictive GT and GC process that might indicate an effective practice Analyse aspects of current protocols that might be relevant for a successful practice	22 individuals undergoing predictive GT 13 for TTR-FAP, 6 for HD, 3 for SCA3	Qualitative	Predictive	Highlight the need of health providers to be armed with personal and professional skills to GC safely and effectively, and provide flexible client-centered care Further training and clinical supervision may help	Education Counseling Support	0.90 (18/20) Data analysis and reflexivity of the account not clearly described
47	Ledo et al. (2013) [50] Portugal	HD TTR-FAP	Compare the behavior symptoms inventory (BSI) psychopathological indices observed before and one year after completion of predictive GT Identify differences between the psychological impact depending on type of risk disease, carrier or non-carrier status and demographic variables (age, gender, marital status) included in the general protocol.	53 individuals who underwent predictive GT (40 for TTR-FAP and 13 for HD) and completed psychological evaluations one year later	Retrospective cohort	Predictive	BSI levels across the time points were higher in both those who tested negative and positive, compared to controls Average BSI levels decreased post-GT regardless of the result, and condition	Counseling Support	1.00 (20/20)
48	Uhrova et al. (2013) [31] Czech Republic	HD	Characterize the differences in psychiatric examination and psychometric measures between people at risk who were recommended to postpone predictive	52 individuals who underwent psychiatric examination as part of the HD predictive GT protocol: 41 continued with GT	Retrospective Cohort	Predictive	Psychiatric examination provides more significant information regarding whether to consider postponing GT than	Education Counseling Support	1.00 (18/18)

			GT, and those who proceeded	(19 tested positive, 22 tested negative), 11 were recommended to postpone			formalized psychiatric screening tools Motivations must be assessed pre-GT, postponing may be required if problematic		
49	Van Rij et al. (2013) [51] The Netherlands	HD	Create a better understanding of the motives and experiences of couples opting for exclusion PND or PGD Study the acceptability of exclusion PGD among candidates	17 couples who had undergone reproductive GT with exclusion 13 PND with exclusion 6 PGD with exclusion (2 couples experienced both)	Qualitative	Reproductive	7 couples had terminated 11 pregnancies, none showed regret. Some elected to have PGD with exclusion to avoid another termination Adequate professional and psychological support required before, during and after PND/PGD	Education Counseling Support	0.95 (19/20) Reflexivity of the account not clearly described
50	Rodrigues et al. (2012) [43] Brazil	SCA3 HD TTR-FAP SCA1 SCA2 SCA6 SCA7	Describe the Brazilian public health system experience of a predictive GT program, run in accordance with the international guidelines for HD, SCAs and TTR-FAP between 1999-2009 Conduct a subsequent survey of the psychological characteristics of individuals who sought predictive GT, to detect differences between groups	183 individuals who commenced predictive GT (147 at risk for SCA3, 22 for HD, 8 for TTR-FAP, 6 for other SCAs) 31 participated in a follow-up interview and 15 had also completed psychological survey pre-GT	Consecutive case series Retrospective cohort Qualitative study not reported	Predictive	Low uptake of post-GT psychological evaluation, reason remains unknown Authors suggest adjustments necessary to provide adequate follow-up	Education Counseling Support	1.00 (18/18) Only quantitative data assessed
51	Alexander et al. (2011) [103] Canada	AD	Assess the effectiveness, outcomes and costs of requesting medical records for the confirmation of client–reported family histories of dementia	275 medical record requests during the 24-month period of January 1, 2005–December 31, 2006	Consecutive case series	Diagnostic	Useful medical records obtained from 92 (33.5%) requests: 77 supported, 15 did not support patient-reported information Patient-reported family history was accurate in 84% Almost 500 hours of GC time spent	Interpretation	0.75 (15/20) Question/ objective, study design and results not sufficiently described Variance estimates not described

	Genetic counseling practice recommended from non-clinical research								
52	Cahn et al. (2020) [98] USA	SCA1, 2, 3, 6, 7, 8, 14, 17, 35	Assess knowledge of genetic risk and perceptions of reproductive options in individuals with a diagnosis of spinocerebellar ataxia	94 individuals with or at risk of SCA, 77 symptomatic	Cross-sectional survey	Reproductive	39.8% would consider PGD, less PND (number unclear) 79.8% would not consider donated embryos, 63-74% would not consider donated gametes Most common decision-making factors: child will not inherit SCA, cost and risk to mother or child	Education	1.00 (18/18)
53	Withers et al. (2019) [69] USA and Mexico	Alzheimer's disease	Examine cultural beliefs about Alzheimer's disease and genetic screening among at risk populations of Mexican heritage	123 individuals from families living in Mexico and California in which Alzheimer's disease mutations were known to run 13 (plus an additional 5) participated in in-depth interviews	Cross-sectional survey Qualitative	Predictive	Few respondents understood their risk of inheriting a pathogenic variant causing Alzheimer's disease in their family Family myths and stigma also present in the family	Education	0.87 (33/38) Method of subject selection, sampling strategy and theoretical framework/ wider body of knowledge not clearly described Reflexivity of the account not described
54	Hagen (2018) [61] Sweden	HD	Explore the intersections between genes, the body and the lived experience of a genetic disease to contribute to a deeper understanding of the lived experience of genetic diseases	11 individuals from HD families 2 affected, 1 presymptomatic carrier, 2 tested negative, 1 untested and at risk, 5 unrelated relatives	Qualitative	Diagnostic Predictive	Lived experience is fluid and dynamic and must be addressed as part of GC process both pre- and post-GT	Counseling Support	0.95 (19/20) Reflexivity of the account not clearly described
55	Hartzfeld et al. (2015) [82] USA	ALS	Learn how familial ALS influences reproductive decisions, the potential influence of others, factors considered during the decision-making process, and participants' overall experience	10 individuals from familial ALS families who were aware of the risk of the disease when they had children	Qualitative	Reproductive	Those who decided to have children always planned on having children, hoped for a cure and compared ALS favourably to other diseases Those who chose not to have children had	Education Counseling	0.80 (16/20) Theoretical framework/ wider body of knowledge not clearly described Conclusions not well supported

			regarding reproductive choices				extensive experience of ALS and caretaking, saw ALS as inevitable tragedy and avoided serious relationships		Reflexivity of account not described
56	Paneque et al. (2015) [48] Portugal	LONDs	Identify quality aspects of effective GC practice in presymptomatic GT for late-onset neurological disorders	45 experts with extensive experience of GC from 11 countries, 29 completed round 1 (64.4%), 13 completed round 2 (31.1%) and 17 competed round 3 (37.7%)	Delphi	Predictive	High quality professional standards; service standards; the consultand-centered practice; and protocol standards were developed Most relevant quality indicators were related to consultand-centered practice, and advanced counseling/ interpersonal skills	Interpretation Education Counseling Support	1.00 (16/16)
57	Hawkins et al. (2013) [47] Canada	HD	Understand the obstacles to GT in terms of accessibility of services in Vancouver, as well as exploring the mechanisms by which this issue may be addressed	33 participants recruited based on a non-probability sample 24 tested, 9 rural, 15 non-rural 9 not tested, 3 rural, 6 non-rural	Qualitative	Predictive	Barriers to accessibility of GT: distance (time and travel, financial and opportunity costs, stress of travel) and inflexibility of the GT process	Interpretation Education Counseling Support	0.75 (15/20) Theoretical framework/ wider body of knowledge, data analysis and reflexivity of the account not clearly described Verification procedures not described
58	Hawkins Virani et al. (2013) [73] Canada	HD	Develop a patient-friendly, comprehensive, accessible Web-based tool to provide accurate information about predictive GT for HD Pilot the content and test usability of the website, and modify the website	33 individuals from HD families, 9 had not had predictive GT, 24 had with 17 tested positive and 7 tested negative 10 individuals who had participated in above study, 5 genetic counselors across North America, 10 HD researchers and	Qualitative Web-based cross-sectional survey	Predictive	Effective website included unbiased overview of important factors to be considered before undergoing predictive GT, interactive diagrams, video documentaries, and personal stories Website considered an effective counseling tool to	Education Counseling Support	0.86 (31/36) Subject characteristics, outcome measures, quantitative analytic methods, theoretical framework/wider body of knowledge and reflexivity of the account not clearly described

				experts and 10 lay individuals invited, 23 completed survey (response rate 65.7%)			support informed decision-making		
59	Etchegary (2011) [25] Canada	HD	Explore the healthcare experiences of families affected by HD, and elicit their suggestions for improvement in the quality of care provided to them.	24 individuals from HD families 2 affected, 3 presymptomatic carriers, 5 tested negative, 2 tested - intermediate result, 2 tested – results not received, 6 untested and at risk, 4 spouses	Qualitative	Diagnostic Predictive	Participants experienced difficulty accessing appropriate healthcare and support, and were frustrated by lack of knowledge of family physicians Regular follow-up and increased education of health professionals recommended	Interpretation Education Support	0.95 (19/20) Reflexivity of the account not clearly described
60	Klitzman (2010) [37] USA	HD (Alpha-1 antitrypsin deficiency, breast cancer)	Investigate the range of possible misunderstandings related to genetics that clients may have, the reasons why these may persist, and the implications that these may have.	21 individuals with or at risk of HD, 15 asymptomatic, 6 symptomatic, 14 had undergone GT, 10 were positive, 10 were negative	Qualitative	Diagnostic Predictive	Participants experienced various misconceptions/ misunderstandings about GT: that they could control disease onset, beliefs about inheriting mutations and physical traits together or that more biological material was inherited from one parent	Education Counseling	0.95 (19/20) Reflexivity of the account not clearly described
61	Schwartz (2010) [93] USA	HD	Explore the unique issues surrounding being diagnosed with a chronic, progressive, genetic disorder through a narrative inquiry approach	10 individuals diagnosed with HD within the past year	Qualitative	Diagnostic Predictive	Lived experience of HD had the following key chapters in the narrative: discovering the existence of HD, confirming the diagnosis, revealing the diagnosis to others, and experiencing the reverberations of HD	Interpretation Education Counseling Support	0.95 (19/20) Reflexivity of the account not clearly described

[#]Conditions not of interest in brackets; *Only population of interest included

KEY: AD: Alzheimer's disease, ALS= amyotrophic lateral sclerosis, CADASIL= cerebral autosomal dominant arteriopathy with subcortical infarcts and leukoencephalopathy, FSHD= Facioscapulohumeral muscular dystrophy, FTD= frontotemporal dementia, GC= genetic counseling, GT= genetic testing, HD= Huntington's disease, PGD= preimplantation genetic diagnosis with in vitro fertilisation, PND= prenatal diagnosis, SCA: spinocerebellar ataxia, TTR-FAP= Familial amyloid polyneuropathy; USA= United States of America